FOR DISCUSSION WITH FACULTY

EdaphNet: where the project stands and what it needs

3 October 2026 Project folder: `edaphnet-v2`

Repository: github.com/Sudeep-Sharma1/edaphnet-v2 (private)

EdaphNet checks soil laboratory reports for mistakes, such as two samples swapped or a value entered ten times too high, by comparing each report with a cheap, independent look at the same soil (its colour in a phone photo). This document summarises what has been done, what the results show, the problems that stand between the work and a paper, and the decisions we need from you.

Short version. The research part is complete as a first study, and a full paper draft exists. The original main claim (colour alone beats the best existing check) **failed** its pre-registered test. The study is still publishable as an honest analysis of *when a cheap check is enough*, likely in a mid-tier applied journal, after about 1–2 weeks of fixes. Its biggest weakness is that the errors were planted and the photo colours were calculated from lab spectra, not real phone photos. Fixing that needs new data, which we need your help to obtain.

---

1. What the project does

Soil sample sent to a lab; a report comes back (organic carbon, nitrogen, total carbon …)

→

Witness a cheap independent observation of the same soil: its colour, or a spectrometer reading

→

Batch check the whole batch of reports is checked together for swaps and a library of error types

→

Result which reports to re-test (5% budget), and what probably went wrong

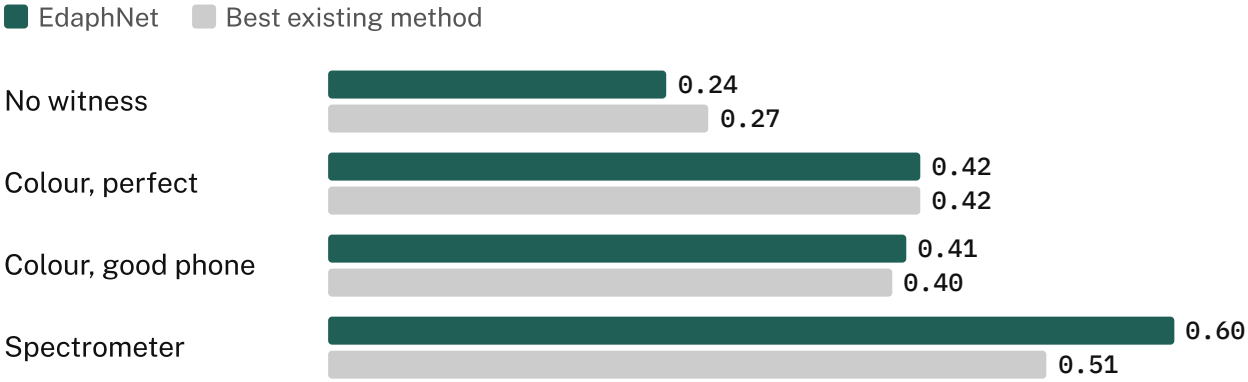
2. Data: what is real and what is simulated

ITEM	REAL OR SIMULATED	DETAILS
Soil samples and lab values	**Real**	23,219 samples: KSSL, USA (19,746 samples, 6,999 locations) and ICRAF-ISRIC, worldwide (3,473). Measured in real laboratories.
Spectra	**Real**	Each sample's Vis-NIR spectrum from a lab instrument (Open Soil Spectral Library, CC-BY).
Errors in reports	Simulated	Swaps, ×10 slips, digit transpositions, unit and column errors planted by our code, so we know where they are. This is standard for measuring detection.
"Phone photo" colour	Simulated	Calculated from each sample's spectrum, plus camera-like noise. A best case: dried, ground soil in perfect light.

3. Main results

All results are the share of erroneous reports found when the top 5% of reports are re-tested. The pass/fail rules were committed before the test data were used.

Pre-registered test on KSSL (doc 21)



Verdict: main claim A1 **failed**: with colour the difference is +0.00 [−0.02, 0.02] (perfect) and +0.01 [−0.01, 0.04] (good phone). With a spectrometer EdaphNet wins by +0.09 [0.07, 0.11]. False-alarm control (A2) and error explanation (A3) passed.

What the overall number hides (exploratory, doc 23)

FINDING	RESULT
Swapped samples found, EdaphNet minus best method (good-phone colour)	**+0.10 [+0.06, +0.14]**
Wrong values found, same comparison	−0.08 [−0.12, −0.05]
Gross analytical errors found, EdaphNet vs best method	0.03 vs 0.26
Correct error type named (spectrometer, the only settings with enough flags to test)	**97–100% vs 24–29%**
Power when false alarms are controlled one batch at a time (colour)	0.19–0.25 at FDR ≈ 0.10

In plain words: colour helps find **swapped samples** but costs some **wrong values**, so the two cancel when errors are half swaps, half wrong values (as we planted them). A spectrometer helps overall. Reusing calibration samples, a design mistake of ours that we found and corrected, silently breaks false-alarm control. That lesson is itself a finding.

---

4. Problems and what we need to do

Ordered by priority. MUST FIX before submission · REVIEWERS WILL ATTACK reduce or answer · HOUSEKEEPING needed but routine.

P1 No real phone photos

REVIEWERS WILL ATTACK | NEEDS FACULTY

Problem

The colour of every sample was calculated from its lab spectrum, not taken from a real phone photo. Real photos (moist soil, shadows, different phones) will be worse.

Why not fixed

Public soil photo sets have soil-type labels but no lab values, and we cannot photograph the US samples. We need photos of samples whose lab values are known.

What to do

- **Public dataset (found):** a Zenodo dataset (CC BY 4.0, published 22 September 2026, DOI 10.5281/zenodo.22876123) has 84 soil samples photographed with four smartphones, with laboratory organic carbon. It allows a real-photo check without new collection. Limits: organic carbon only; soil location and use of a colour card not stated on the record page. Not yet downloaded (6 GB; one phone's photos ≈1.5 GB).
- **Indian pilot:** 30–50 already-tested soil samples, each photographed with a phone and a colour card, following the protocol already written (doc 15). Proposed partner: the soil science laboratory of VIT's School of Agricultural Innovations and Advanced Learning (VAIAL), Vellore. About 2–4 weeks; a colour card costs roughly ₹2,000–5,000 (estimate).

Need from you

Approval and an introduction to the VIT VAIAL soil science faculty (or another laboratory you know). A draft request email is ready.

P2 No real errors

REVIEWERS WILL ATTACK | NEEDS FACULTY

Problem	All errors were planted by our code. We do not know how often real labs swap samples or mistype values, and the overall verdict depends on that mix.
Why not fixed	With ordinary reports nobody knows which ones are wrong. Finding out needs **blind duplicates** (the same soil sent twice under different names) or a lab's records of re-tests. Real errors are rare (perhaps 1–5%), so hundreds of samples are needed.
What to do	For this paper: state it clearly as the main limitation (already in the draft). For a second paper: blind-duplicate study with a partner lab (Step 6). Also ask whether proficiency-testing schemes (e.g. WEPAL, NAPT) or a lab's correction records can be shared; we have not yet checked whether such data are obtainable.
Need from you	A lab contact and a budget decision for Step 6.

P3 Unexplained gross-error result

MUST FIX | STUDENT

Problem	EdaphNet finds only 0.03 of gross analytical errors versus 0.26 for the best method. A second review also reported that three different EdaphNet variants give exactly the same value (0.034); this still has to be checked against our run records.
Why it matters	It could be a real limitation (the rule-based method uses "organic carbon ≤ total carbon", which our model cannot express) or a scoring problem. If it is a bug, other numbers may change.
What to do	Pull the gross-error cases from the saved predictions, inspect their scores, and test the explanation. About half a day. This comes first.

P4 The error mix decides the winner

REVIEWERS WILL ATTACK | STUDENT

Problem	We planted 50% swaps and 50% wrong values. That choice is arbitrary, and with colour the overall result depends on it.
What to do	Add a figure showing which method wins as the share of swaps goes from 10% to 90%, by re-scoring the saved predictions. Labelled exploratory. This turns a weakness into a practical guide for labs: "if your errors are mostly swaps, use this".

P5 "Why such a complex model?"

REVIEWERS WILL ATTACK | STUDENT

Problem	Some parts of the method do not help: the error library costs swaps (−0.07) and the structured swap prior has no effect. A simple swap matcher may come close on swaps alone (to be confirmed from the records).
What to do	Present the model's real advantage, **explanation** (naming the correct error type 97–100% vs 24–29%), as a main result, and report the ablations openly. Note honestly that this was measurable only with a spectrometer.

P6 Only one main dataset, and no Indian soils

REVIEWERS WILL ATTACK | STUDENT + FACULTY

Problem	The deciding test used KSSL (USA, carbon-related properties only). The ICRAF test was affected by our calibration mistake. None of the tested soils is known to be Indian, although India (large public soil-testing programmes) is the motivation.
What to do	Request LUCAS (about 40,000 European samples, free on request) and run a confirmation test, pre-registered first. For Indian soils: the VIT pilot (P1), and possibly the authors of a 2025 smartphone soil-carbon study in the Indo-Gangetic Plains (data availability unknown).
Need from you	Whether the paper must include Indian data, or India as motivation is acceptable. The LUCAS request may need an institutional email or supervisor name.

P7 Weak evidence on unknown error types

MUST FIX | STUDENT

Problem	The test with error types the model does not know used only one random seed, so its intervals are wide.
What to do	Rerun with 5 seeds. Mostly computer time on the laptop (about a day).

P8 Draft corrections

MUST FIX STUDENT

What to do

- Abstract: mark the swap/value numbers as exploratory.
- Replace "needs thousands of distinct verified reports" with the accurate statement (calibration size relative to pool size).
- State that the best existing method keeps more power with colour at every pool size.
- Fix overlapping labels in Figure 4.
- Add a method diagram.

P9 Literature search, reproducibility and disclosures

HOUSEKEEPING STUDENT + FACULTY

What to do

- **Formal search** in Scopus or Web of Science through the library, logged, to back the "no prior work" claim, and check every citation marked [verify].
- **Reproducibility:** a data-download script so a fresh copy can reproduce the results, and a frozen release with a DOI on Zenodo.
- **AI-assistance statement:** the code and drafts were developed with an AI assistant (Claude); journals require this to be declared.
- **Patent:** the repository was public for a period before being made private (dates are in the GitHub history). Discuss with the IP office before any preprint.

P10 App layer kept separate

HOUSEKEEPING STUDENT

Status

A separate farmer-facing app exists: leaf-disease photo check, crop suggestions from soil and climate, soil-colour estimate and report check, with a camera page. It is not part of this paper.

Limits

- The leaf model uses standard pretrained features with a simple classifier, not a custom model; its field-photo accuracy is still being measured.
- Crop rankings use FAO EcoCrop ranges and have not been checked by an agronomist.
- Soil-type prediction is on hold for lack of an audited image set.

5. Proposed plan

PHASE	WORK	TIME
A. Fix the paper	P3 gross-error check, then P7 five seeds, P8 draft corrections, P4 error-mix figure, method diagram, reproducibility script	1–2 weeks
B. Faculty actions	Review of the reframed claim, formal literature search, LUCAS request, IP office, journal choice	in parallel
C. Real-photo check	Zenodo 84-sample smartphone dataset (P1), and/or 30–50 Indian samples photographed at VIT VAIAL; added to this paper if done in time	1–4 weeks
D. Second paper	Real errors through blind duplicates with a partner lab (P2), plus LUCAS confirmation	2–4 months

6. Decisions we need from you

1. **Submit now or add a pilot?** Submit with planted errors and calculated colour (limitation stated), or wait about a month to add 30–50 real photos?
2. **Indian data.** Must the paper include Indian soils, or is India acceptable as the motivation with testing on US and global data?
3. **Soil samples with lab values.** Can you introduce me to the VIT VAIAL soil science laboratory (or another lab) for 30–50 already-tested samples? May I download the public Zenodo dataset for a real-photo check?
4. **Framing.** Do you agree with presenting the work as a study of when a cheap witness is enough, rather than a new-method paper?
5. **Target journal.** Suggested: SOIL, European Journal of Soil Science, or Computers and Electronics in Agriculture.
6. **Patent, authorship and AI disclosure.** Whether to approach the IP office before a preprint (the repository was public for a period), who will be co-authors, and the wording of the AI-assistance statement.
7. **Step 6 budget and lab partner** for the blind-duplicate study.

Every number in this document comes from the project's committed documents and run records: doc 21 (pre-registered KSSL results), doc 22 (progress report), doc 23 (exploratory analyses), doc 15 (data-collection protocol), doc 24–25 (app layer). The Zenodo dataset details are from its record page (accessed 3 October

2026). Items described as "to be checked" have not yet been verified against the records. This document was prepared with the help of an AI assistant (Claude).